Experiment-7

Write a Servlet program to store a user's preferences (like theme or language) using cookies. Retrieve and display these preferences on subsequent visits.

AIM:

To develop a Servlet-based web application that stores user preferences (such as theme or language) using cookies and retrieves them on subsequent visits to provide a personalized user experience.

ALGORITHM:

1. Input:

- The user selects preferences such as theme (light/dark) and language from a web form.

2. Steps for Implementing Cookies in Servlet:

Step 1: Create the Preference Form (HTML)

- Design an HTML form with fields for selecting user preferences (e.g., theme and language).
- Provide a submit button to send the data to the servlet.

Step 2: Create the Preference Servlet

- Write a Java servlet to handle the form submission.
- Retrieve user input using `request.getParameter()` method.

Step 3: Create Cookies

- Create cookies for each preference using:
  `Cookie cookie = new Cookie("name", "value");`
- Example: theme and language cookies.

Step 4: Add Cookies to Response

- Add cookies to the response using:
  `response.addCookie(cookie);`
- Optionally, set cookie expiry time using:
  `cookie.setMaxAge(time_in_seconds);`

Step 5: Retrieve Cookies on Subsequent Visits

- Use `request.getCookies()` to get all cookies.
- Loop through cookies and retrieve stored values based on cookie names.

Step 6: Display User Preferences

- Display the retrieved preferences (theme/language) on the webpage.
- Apply the selected theme or language settings accordingly.

Step 7: Handle Default Values

- If no cookies are found, display default preferences.

CODE:
PreferenceServlet.java:

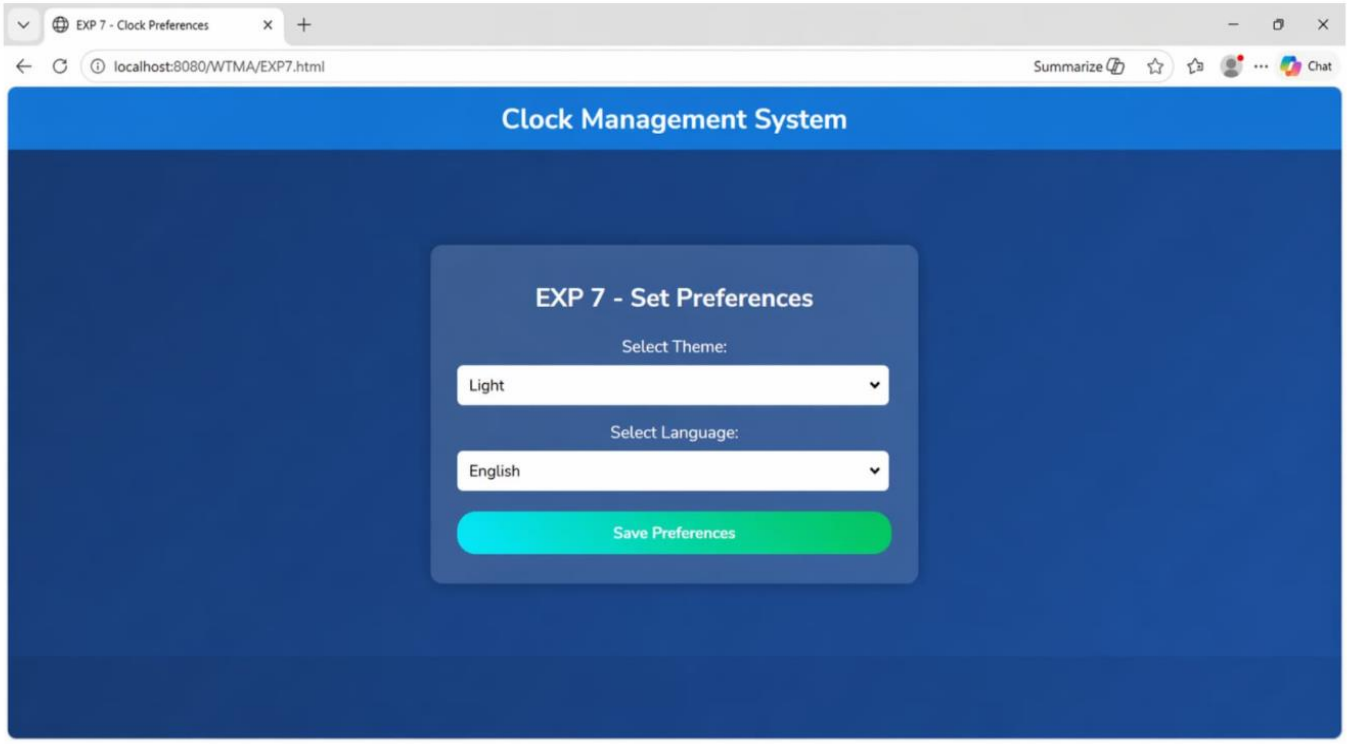
```
package WTMA;


import java.io.*;

import jakarta.servlet.ServletException;

import jakarta.servlet.http.*;

import jakarta.servlet.annotation.*;

@WebServlet("/Preference")

public class PreferenceServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    public void doGet(HttpServletRequest request, HttpServletResponse response)

            throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter pw = response.getWriter();

        String theme = "light";

        String language = "English";

        Cookie[] cookies = request.getCookies();

        if (cookies != null) {

            for (Cookie c : cookies) {

                if (c.getName().equals("theme")) theme = c.getValue();

                if (c.getName().equals("language")) language = c.getValue();

            }

        }

        boolean isDark = theme.equals("dark");
```

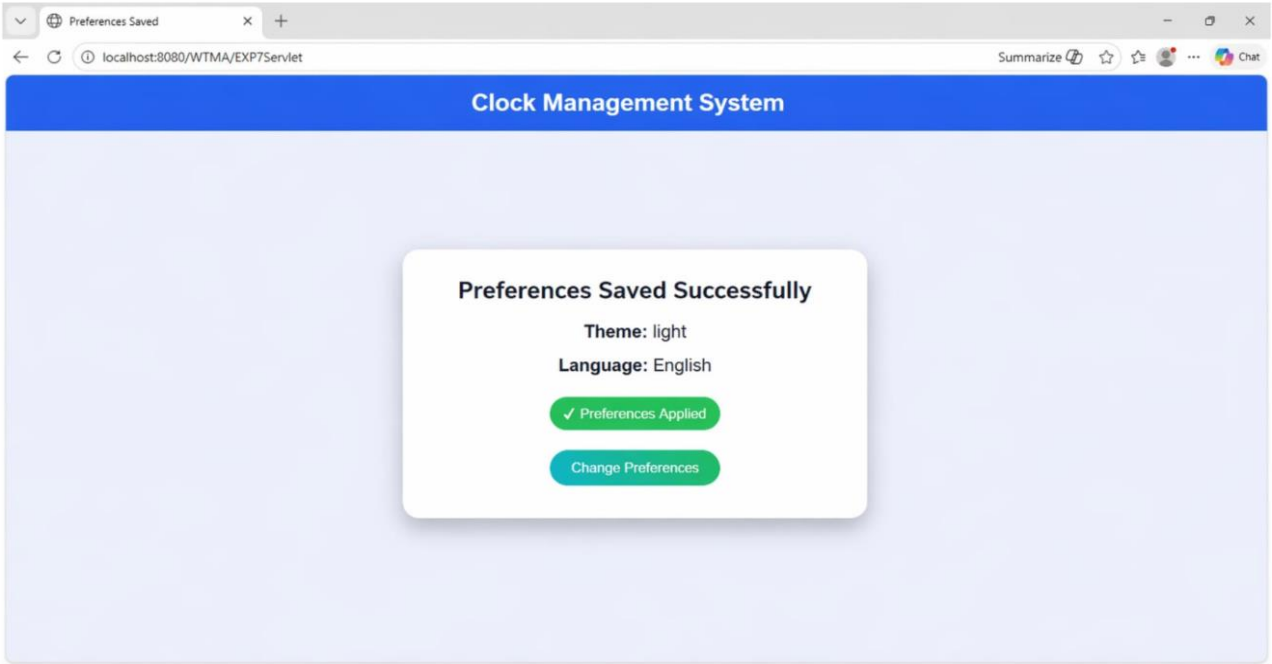
```
String pageBg = isDark ? "#0f172a" : "#e0e7ff";

String cardBg = isDark ? "#1e293b" : "#ffffff";

String textColor = isDark ? "#ffffff" : "#1f2937";

String subText = isDark ? "#cbd5e1" : "#6b7280";

pw.println("<html>");

pw.println("<head><title>School Portal</title>");

pw.println("<style>");

pw.println("body { margin:0; font-family:Segoe UI; background:" + pageBg + "; color:" + textColor + "; }");

pw.println(".wrapper { display:flex; justify-content:center; align-items:center; height:100vh; }");

pw.println(".card { width:450px; background:" + cardBg + "; padding:25px; border-radius:15px; box-shadow:0 10px 30px rgba(0,0,0,0.3); }");

pw.println(".title { font-size:22px; font-weight:bold; text-align:center; margin-bottom:5px; }");

pw.println(".subtitle { text-align:center; color:" + subText + "; margin-bottom:20px; }");

pw.println(".infoBox { background:rgba(0,0,0,0.05); padding:12px; border-radius:10px; margin-bottom:20px; }");

pw.println(".row { margin:12px 0; }");

pw.println("select { width:100%; padding:12px; margin-top:5px; border-radius:8px; border:1px solid #ccc; }");

pw.println(".btn { width:100%; padding:12px; background:linear-gradient(90deg,#4f46e5,#06b6d4); color:white; border:none; border-radius:8px; cursor:pointer; font-size:15px; }");

pw.println(".btn:hover { opacity:0.9; }");

pw.println("</style>");

pw.println("</head>");

pw.println("<body>");

pw.println("<div class='wrapper'>");

pw.println("<div class='card'>");

pw.println("<div class='title'>□ School Management Portal</div>");

pw.println("<div class='subtitle'>User Preference Dashboard</div>");

pw.println("<div class='infoBox'>");

pw.println("□ <b>Theme:</b> " + theme + "<br>");
```

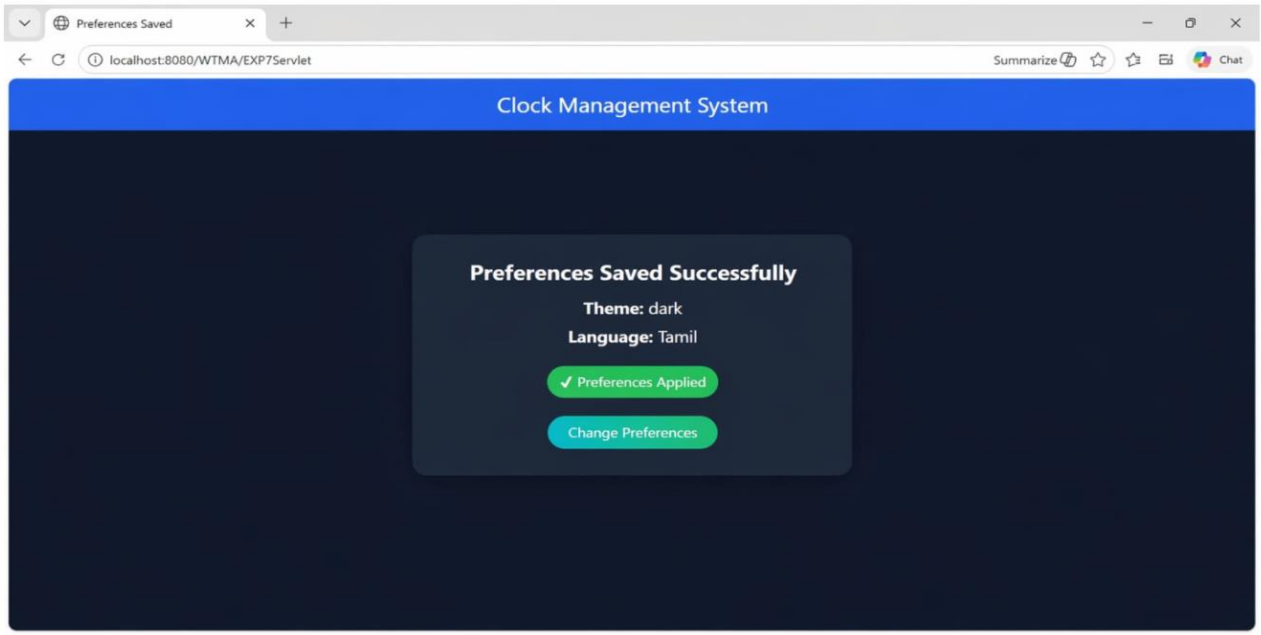
```
    pw.println("  <b>Language:</b> " + language);
    pw.println("</div>");
    pw.println("<form method='post' action='./Preference'>");
    pw.println("<div class='row'>");
    pw.println("  Select Theme");
    pw.println("<select name='theme'>");
    pw.println("<option value='light'" + (theme.equals("light") ? "selected" : "") + ">Light Mode</option>");
    pw.println("<option value='dark'" + (theme.equals("dark") ? "selected" : "") + ">Dark Mode</option>");
    pw.println("</select>");
    pw.println("</div>");
    pw.println("<div class='row'>");
    pw.println("  Select Language");
    pw.println("<select  name='language'>");
    pw.println("<option value='English'" + (language.equals("English") ? "selected" : "") + ">English</option>");
    pw.println("<option value='Tamil'" + (language.equals("Tamil") ? "selected" : "") + ">Tamil</option>");
    pw.println("<option value='Hindi'" + (language.equals("Hindi") ? "selected" : "") + ">Hindi</option>");
    pw.println("</select>");
    pw.println("</div>");
    pw.println("<br>");
    pw.println("<button class='btn' type='submit'>  Save Preferences</button>");
    pw.println("</form>");
    pw.println("</div>");
    pw.println("</div>");
    pw.println("</body></html>");
  }
  public void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
```

```
        String theme = request.getParameter("theme");

        String language = request.getParameter("language");

        Cookie themeCookie = new Cookie("theme", theme);

        Cookie langCookie = new Cookie("language", language);

        themeCookie.setMaxAge(60 * 60 * 24 * 7);

        langCookie.setMaxAge(60 * 60 * 24 * 7)

        response.addCookie(themeCookie);

        response.addCookie(langCookie);

        response.sendRedirect("./Preference");

    }

}
```

OUTPUT:




Preferences Saved
localhost:8080/WTMA/EXP7Servlet
Summarize
Chat
Clock Management System
Preferences Saved Successfully
Theme: light
Language: English
✓ Preferences Applied
Change Preferences


Preferences Saved
localhost:8080/WTMA/EXP7Servlet
Summarize
Chat
Clock Management System
Preferences Saved Successfully
Theme: dark
Language: Tamil
✓ Preferences Applied
Change Preferences

Result:

The Servlet program was successfully implemented to store user preferences using cookies. The application was able to retrieve and display the stored preferences on subsequent visits, providing a personalized experience to the user. Cookies were effectively used to maintain state information across multiple requests.